MSc in Finance

FBA1029[1] Dissertation

The AI-ESG Paradox: Evaluating the Environmental Cost of AI-Driven Sustainable Investing

Anh Vu

by

Suyash Sharad Sawant
(A00049679)
suyashrch@gmail.com

Supervised by

Julie Byrne

DCU BUSINESS SCHOOL

2025-26

Word count: 6,224

Abstract

Sustainable investing in Europe has become a multi-trillion-euro business, and over the same years the asset managers and rating agencies behind it have leaned more and more on machine learning and large-scale data processing. That raises an awkward question. If the data and AI systems used to label funds as sustainable have a carbon cost of their own, and it is not clear that investors benefit from them, is part of the ESG industry working against its own goal? I call this the AI-ESG paradox and test it in two ways. The first uses industry figures from EFAMA's Asset Management in Europe reports, MSCI's disclosures and the IEA. The second uses a Bloomberg panel of 3,917 equity funds domiciled in Europe (2,456 classified under SFDR Article 8 or 9 and 1,461 under Article 6), with monthly data from 2019 to 2024, an AI Intensity Index built from the funds' own descriptions, and the emissions of the five largest listed ESG-data providers.

The ESG-data business is growing much faster than the money it serves. MSCI's Sustainability and Climate revenue went from USD 91m in 2019 to USD 327m in 2024, about 29% a year, while European assets under management grew by 5.0% a year. Investors, on the other hand, did not get better returns for it. After adjusting for risk, Article 8 and 9 funds performed in line with conventional funds (a gap of -0.29% a year, nowhere near significant), and the funds that look most 'AI-intensive' only appear to do better because many of them simply invest in AI companies. On emissions, all five providers became more carbon-efficient, with the median cutting its intensity by 59%, yet the emissions that can be attributed to MSCI's ESG business still rose by 60%, because the business grew faster than it became efficient. Investors also clearly bought the label: Article 9 funds attracted strong inflows until 2021 and then lost money steadily from 2022. At country level, ESG fund assets grow roughly in line with each national market and make up a bigger share of assets where markets are deeper. The numbers involved are small next to the emissions of the companies these funds own, so my argument is about transparency: the footprint of the ESG-data supply chain should be part of what ESG disclosure covers. All the results are associations, not proof of cause and effect.

JEL codes: G11, G23, G14, Q56, O33

Keywords: ESG investing; SFDR; mutual fund performance; fund flows; artificial intelligence; ESG data; carbon emissions

Contents

Abstract

1. Introduction and Motivation

2. Literature Review and Theoretical Framework

3. Research Questions and Hypotheses

4. Data Sources and Construction

5. Methodology

6. Results I: Industry and Country Evidence

7. Results II: Fund-Level Evidence

8. Robustness and Endogeneity

9. Discussion and Limitations

10. Conclusion and Future Research

References

Appendix A: Data and Code Availability

Appendix B: Variable Definitions

Appendix C: AI Intensity Index Dictionary

List of tables

Table 4.1 Data sources

Table 4.2 Country sample (end-2024)

Table 4.3 Fund sample

Table 6.1 H1: market depth and the Article 9 share

Table 6.2 H2: scaling of ESG fund assets

Table 6.3 H3: European AuM and the ESG-data layer

Table 6.4 H4: asset allocation

Table 6.5 IEA context

Table 7.1 Four-factor alphas by SFDR group

Table 7.2 H5: AI intensity and alpha

Table 7.3 H6: ESG and conventional funds

Table 7.4 H7: investor flows

Table 7.5 H8: MSCI emissions and its ESG business

Table 7.6 H8: provider emissions intensity

Table 8.1 Robustness tests

Table B.1 Variable definitions

Table C.1 AI Intensity Index dictionary

List of figures

Figure 6.1 Market depth and the Article 9 share

Figure 6.2 SFDR fund assets by country

Figure 6.3 European AuM and the ESG-data layer

Figure 6.4 European asset allocation
Figure 7.1 Cumulative returns: ESG and conventional funds
Figure 7.2 Distribution of alphas
Figure 7.3 Investor flows by SFDR group
Figure 7.4 Emissions of the ESG-data layer

1. Introduction and Motivation

1.1 Background

Europe is where sustainable investing has gone furthest. Since 10 March 2021, the Sustainable Finance Disclosure Regulation (SFDR) has asked every fund sold in the EU to place itself in one of three groups: Article 6 funds make no sustainability claim, Article 8 funds promote environmental or social characteristics, and Article 9 funds have sustainable investment as their objective. By the end of 2024, EFAMA estimated that more than EUR 4 trillion sat in Article 8 and 9 funds managed in the countries it covers (EFAMA, 2025).

Over the same period, the industry has invested heavily in data and AI. Asset managers use natural language processing to read company reports and news, rating agencies use machine-learning models to produce and update ESG scores, and many funds use quantitative screens to build and rebalance portfolios. None of this runs for free in environmental terms. Strubell, Ganesh and McCallum (2019) estimated that training one large language model with neural architecture search emitted around 626,000 lbs of CO2-equivalent, roughly 284 tonnes, and Patterson et al. (2021) showed that running models day to day adds to that. Across the whole economy, the IEA (2025) puts data-centre emissions at about 180 Mt of CO2 in 2024, or 0.5% of global fuel-combustion emissions, and expects that share to reach 1.0-1.4% by 2030 as AI use grows.

This is the tension behind the dissertation, which I refer to as the **AI-ESG paradox**. The tools used to find, rate and sell sustainable funds may be a growing and largely invisible source of the emissions those funds are meant to help reduce, and it is not obvious that the investors paying for these tools get anything back. The paradox has a cost side and a benefit side, and I look at both: whether the data and AI layer behind ESG investing is growing faster than the assets it serves, with a rising footprint, and whether more data- or AI-intensive funds actually do better for their investors.

1.2 Research questions

- **RQ1.** How do the size and depth of national asset-management markets relate to how much money sits in SFDR-labelled funds, and to the split between Article 8 and Article 9?
- **RQ2.** Has the ESG-data layer, using MSCI as the example, grown faster than the European asset-management industry it serves?
- **RQ3.** Do AI- or data-intensive ESG funds, and ESG funds in general, earn higher risk-adjusted returns than similar funds?
- **RQ4.** How do investor flows respond to ESG labels and to AI intensity, and did that change after 2021?
- **RQ5.** How big is the emissions footprint of the ESG-data layer, and is it growing or shrinking?

1.3 Contribution

Research on ESG fund performance and research on the environmental cost of AI have mostly developed separately, and this study brings them together. Along the way it does three things that, as far as I can tell, have not been done before for Europe: it builds a transparent, text-based AI Intensity Index for 2,456 ESG equity funds; it compares performance and flows for 3,917 funds across all three SFDR groups through the 2022 energy and interest-rate shock; and it estimates the emissions that belong to a leading ESG-data business using that company's own figures.

1.4 Structure

Chapter 2 reviews the literature and Chapter 3 sets out the hypotheses. Chapter 4 describes the data, including how the AI Intensity Index was built, and Chapter 5 the methods. Chapters 6 and 7 present the industry and fund results, Chapter 8 tests how robust they are, Chapter 9 discusses what they mean and where they fall short, and Chapter 10 concludes.

2. Literature Review and Theoretical Framework

2.1 ESG investing, performance and flows

Friede, Busch and Bassen (2015) pulled together more than 2,000 studies and found that most report a positive or neutral link between companies' ESG performance and their financial results. Evidence on funds is less flattering. Renneboog, Ter Horst and Zhang (2008) found that socially responsible funds in most countries trail conventional funds slightly after adjusting for risk, usually by amounts that are not statistically significant. Theory offers a reason. Pedersen, Fitzgibbons and Pomorski (2021) and Pástor, Stambaugh and Taylor (2021) show that if investors like holding green assets, those assets should offer lower expected returns in equilibrium, but they can still beat the market for long stretches when concern about climate change rises faster than expected. Pástor, Stambaugh and Taylor (2022) find exactly that in the strong green returns of 2012-2020. This matters here, because my sample runs straight through 2022, when energy stocks soared and interest rates rose sharply.

Investors clearly care about sustainability, and not only for financial reasons. Hartzmark and Sussman (2019) show that when Morningstar introduced sustainability ratings, money flowed into the highest-rated funds and out of the lowest-rated ones. Riedl and Smeets (2017) find that social preferences and signalling explain more of why people hold responsible funds than return expectations do. At the same time, flows chase past performance (Sirri and Tufano, 1998), so any study of flows needs to control for returns, as this one does.

2.2 How reliable are ESG labels and data?

Berg, Kölbel and Rigobon (2022) show that the main ESG rating agencies often disagree about the same company, because they measure different things, measure them differently and weight them differently. Kotsantonis and Serafeim (2019) point to patchy reporting standards and heavy reliance on unverified company disclosures, and Chatterji et al. (2016) find the disagreement is worst for environmental scores. Raghunandan and Rajgopal (2022) find that ESG funds do not reliably hold lower-emitting companies. All of this work looks at the companies inside the portfolio. I look one step back, at the data supply chain that produces the label in the first place.

2.3 AI in asset management

Machine learning is now used at almost every stage of investing, from forecasting returns to executing trades (López de Prado, 2018). Fisch, Labouré and Turner (2019) show that automated advice can cut costs for investors, and AI-driven ESG data firms such as MSCI, Clarity AI and Arabesque S-Ray now sit behind the screening of most European ESG funds. Whether funds that lean heavily on AI actually earn better risk-adjusted returns is still an open question, and any benefit has to be large enough to cover the extra data and technology costs that end up in fees.

2.4 The environmental cost of AI

The best-known estimate of what training a model costs in carbon comes from Strubell, Ganesh and McCallum (2019). Patterson et al. (2021) show that the figure depends heavily on the hardware and on how clean the local electricity grid is, Lacoste et al. (2019) built a calculator for training emissions, and Schwartz et al. (2020) argue that researchers should report efficiency alongside accuracy, an idea

they call 'Green AI'. Most of this work is about training. The constant, repeated use of models, which is what an ESG-data platform does when it re-scores thousands of companies, has had much less attention. Looking ahead, the IEA (2025) expects data-centre emissions to grow about 2.5 times by 2030 in its Lift-Off Case. It also estimates that wider use of existing AI tools in energy, industry and buildings could cut up to 1,400 Mt of emissions in 2035, roughly three times projected data-centre emissions, but says this is not currently on track.

2.5 Theoretical framework

Three ideas shape how I approach the question. **Stakeholder theory** (Freeman, 1984) suggests that a fund's environmental footprint should include what its own operations and suppliers emit, not just its holdings. **Signalling theory** (Spence, 1973) explains why investors lean on visible markers such as SFDR labels, ratings and claims of technological sophistication when the underlying quality is hard to judge, and, as in Akerlof's (1970) market for lemons, why weaker products can then pass for stronger ones. The **SASB materiality framework**, now part of the International Sustainability Standards Board, is the reason I use a company that reports under SASB as the main case study. Between them, these ideas suggest that labels and claims of sophistication will attract money whether or not they improve outcomes, and that the costs of the data layer will go unnoticed unless someone discloses them.

2.6 Research gap

I have not found a study that compares the growth of the ESG-data business with the assets it serves, tests whether AI or data intensity helps European ESG funds perform across all three SFDR groups, or measures the emissions that belong to an ESG-data business. This dissertation tries to do all three.

3. Research Questions and Hypotheses

I test eight hypotheses. The first four use industry and country data and are covered in Chapter 6; the other four use fund and provider data and are covered in Chapter 7.

H1 (RQ1): Market depth and the 'dark green' share. EU countries with deeper asset-management markets, measured as assets under management relative to GDP, hold a larger share of their SFDR-labelled assets in Article 9 funds.

H2 (RQ1): Scaling. Article 8 and 9 fund assets grow in proportion to the size of each country's asset-management industry, so the elasticity is one.

H3 (RQ2): The ESG-data layer outgrows the industry. MSCI's Sustainability & Climate business grows much faster than European assets under management.

H4 (context): Portfolio mix. European portfolios moved from bonds towards equities, which increases their direct exposure to company emissions.

H5 (RQ3): AI intensity and performance. Among Article 8 and 9 funds, AI or data intensity has no link with risk-adjusted performance once costs, size and style are accounted for. The alternatives are a positive effect, if AI gives an information edge, or a negative one, if its cost outweighs the benefit.

H6 (RQ3): ESG and performance. Article 8 and 9 funds earn the same risk-adjusted returns as conventional Article 6 funds.

H7 (RQ4): Flows. Investors put more money into funds with stronger ESG labels, Article 9 more than Article 8, after allowing for past performance, and this preference weakened after 2021.

H8 (RQ5): The footprint of the ESG-data layer. Emissions linked to providing ESG data are rising in absolute terms, even if providers are becoming less carbon-intensive.

4. Data Sources and Construction

4.1 Sources

Every figure in this dissertation comes either from a public source that I checked against the original document, or from data I downloaded myself from a Bloomberg terminal in the DCU Bloomberg lab on 2 November 2026. Nothing has been estimated, filled in or interpolated. Table 4.1 lists the sources.

Table 4.1: Data sources

Dataset	Content	Coverage	Source
Industry and country statistics	AuM, AuM/GDP, SFDR Article 8/9 assets, asset allocation, AuM over time	EU/European countries, 2018-2024	EFAMA (2023, 2025), checked
Fund universe and characteristics	Type, domicile, launch date, SFDR group, size, fees, manager, benchmark, description	3,917 equity funds	Bloomberg
Fund returns and assets	Monthly total return with dividends reinvested, NAV, total assets, in EUR	Jan 2019 - Dec 2024	Bloomberg
Risk factors	Developed-market Fama-French three factors, momentum, risk-free rate	Jan 2019 - Dec 2024	Kenneth R. French Data Library
ESG-data providers	Scope 1-3 emissions, energy, staff, revenue: MSCI, S&P Global, Moody's, LSEG, Morningstar	2019-2024	Bloomberg
MSCI operations	Revenue by segment; headcount and staff turnover	2019-2024	Bloomberg; MSCI SASB tables, checked
Climate context	Data-centre emissions and scenarios; global CO2	2024, 2030, 2035	IEA (2025); Our World in Data (2025), checked

Notes: The data files and the analysis code are available on request (Appendix A).

4.2 Industry and country data

The country figures come from the 17th edition of EFAMA's Asset Management in Europe (EFAMA, 2025). Its statistical annex gives each country's assets under management and AuM/GDP at the end of 2024, and Exhibit 4.9 gives Article 8 and 9 fund assets by country. I kept only EU member states, since SFDR does not apply elsewhere. That leaves fifteen countries with Article 8 figures and AuM/GDP, nine of which also report Article 9 assets (Table 4.2). The European AuM series for 2019-2024 is taken from EFAMA (2025, Exhibit 1.1). Asset allocation comes from EFAMA (2023, Exhibit 4.2) for 2018-2022 and EFAMA (2025, Exhibit 4.2) for 2023-2024, and the two editions give the same numbers for the years they share.

Table 4.2: Country sample: asset-management scale and SFDR-labelled fund assets, end-2024

Country	Total AuM (EUR bn)	AuM / GDP (x)	Article 8 (EUR bn)	Article 9 (EUR bn)	Article 9 share
France	5,290	1.81	1,552	95	5.8%
Netherlands	2,119	1.87	717	15	2.0%
Italy	1,528	0.70	229	12	5.0%
Spain	583	0.37	144	n.r.	n.r.
Denmark	548	1.38	485	20	4.0%
Belgium	447	0.73	181	1	0.5%
Austria	202	0.42	112	2	1.8%
Poland	89	0.11	5	0.1	2.0%
Czech Republic	71	0.22	3	n.r.	n.r.
Hungary	57	0.28	5	0.02	0.4%
Portugal	52	0.18	15	0.3	2.0%
Greece	30	0.12	1	n.r.	n.r.
Slovakia	12	0.09	1	n.r.	n.r.
Slovenia	9	0.13	3	n.r.	n.r.
Croatia	5	0.05	1	n.r.	n.r.

Notes: EU member states that appear in both EFAMA (2025) Annex A.1 and Exhibit 4.9. AuM/GDP is a ratio (1.81 = assets equal to 181% of GDP). n.r. = Article 9 assets not reported by EFAMA for that country; these countries are used in H2 (Article 8 only) but not in H1. Turkey, Switzerland, the UK and Norway are excluded because SFDR does not apply or data are incomplete. Sources: EFAMA, Asset Management in Europe (2023, 2025 editions); MSCI SASB tables; Bloomberg.

4.3 Fund sample

I built the fund universe with Bloomberg's fund screener. The screen keeps open-end funds and ETFs (primary share class only) that invest in equities, are domiciled in one of ten EU countries (Ireland, Luxembourg, France, Germany, the Netherlands, Sweden, Spain, Italy, Belgium or Denmark), are still active, were launched before 1 January 2019 and hold more than EUR 50 million, with all assets converted to euros. I ran the same screen twice, once for Article 8 and 9 funds (2,456 funds: 2,298 Article 8 and 158 Article 9) and once for Article 6 funds (1,461). I then downloaded monthly total returns, net asset values and fund assets from January 2019 to December 2024, which gives 280,270 fund-month returns. As a check, the returns line up closely with month-to-month changes in NAV (correlation 0.996). Twelve monthly returns larger than 50% in either direction, all caused by share splits or mergers, were removed. Net flows are calculated in the usual way (Sirri and Tufano, 1998) as Flow(t) = [TNA(t) - TNA(t-1) x (1 + R(t))] / TNA(t-1), and the top and bottom 1% are capped.

Table 4.3: Fund sample by SFDR classification, 2019-2024

Group	Funds	Median AuM (EUR m)	ETF share	Luxembourg / Ireland	Median TER (%)	Data-intensive (All > 0)	Mean annual return	Mean volatility
Article 6	1,461	340	36%	35% / 33%	n.a.	n.a.	10.0%	17.5%
Article 8	2,298	394	7%	46% / 15%	1.49	5.7%	9.9%	16.8%
Article 9	158	361	1%	53% / 7%	1.68	4.4%	9.9%	16.9%

Notes: Open-end funds and ETFs; primary share class; equity focus; domiciled in ten EU countries; incepted before 2019; assets above EUR 50m at the screening date. Returns are total returns in EUR (dividends reinvested), monthly, January 2019 to December 2024. Expense ratios and fund descriptions were collected for Article 8/9 funds only. Source: Bloomberg fund screen, 2 November 2026.

One limitation is worth flagging now. Because the screen only finds funds that still existed in November 2026, funds that closed or merged during 2019-2024 are missing. The same screen was used for all three groups, so this affects the overall level of performance more than the comparison between groups.

4.4 The AI Intensity Index

No database records how much a fund relies on AI, so I built a measure from text, an approach that is now common in finance research. The AI Intensity Index (AII) scores each Article 8 and 9 fund's name and Bloomberg description against a three-level word list (Appendix C). Explicit AI and machine-learning terms carry a weight of 3, quantitative or algorithmic process terms a weight of 2, and rules-based, factor or automation terms a weight of 1. The weights present are added up and scaled to run from 0 to 10. In total, 138 of the 2,456 ESG funds score above zero. I also use two simpler yes/no versions: 'AI or quantitative process' and 'explicit AI or machine-learning wording'. One distinction turned out to matter a great deal: some funds invest in the AI theme, such as robotics or AI-company funds, which is very different from using AI to manage money. I flag these 21 AI-themed funds separately.

4.5 ESG-data provider data

For the five largest listed ESG-data and ratings providers (MSCI, S&P Global, Moody's, LSEG and Morningstar), I downloaded Scope 1, Scope 2 (market-based) and Scope 3 emissions, energy use, staff numbers and revenue for 2019-2024 from Bloomberg. MSCI's revenue by segment, including its Sustainability & Climate segment (called 'ESG and Climate' before 2023), also comes from Bloomberg, and its headcount from its SASB reports.

5. Methodology

5.1 Country tests (H1, H2, H4)

$$(1)\quad Art9Share_c = a + b \times (AuM/GDP)_c + e_c$$

$$(2)\quad ln(ESG\ AuM_c) = a + h \times ln(Total\ AuM_c) + e_c\ ,\quad H0: h = 1$$

Equation (1) is a simple cross-country regression. Because there are only nine countries, I also report Pearson and Spearman correlations, bootstrap the slope 5,000 times and re-run the test leaving out one country at a time. Equation (2) estimates how ESG fund assets scale with the size of the whole industry and tests whether they scale one-for-one. For H4, I regress each asset class's share of European AuM on a time trend.

5.2 Growth of the ESG-data layer (H3)

I index European AuM, MSCI's total revenue, its Sustainability & Climate revenue and its headcount to 2019 = 100 and compare their annual growth rates. With six years of data this is a descriptive comparison rather than a formal test.

5.3 Fund performance (H5, H6)

$$(3)\quad R_it - Rf_t = a_i + b1\ MKT_t + b2\ SMB_t + b3\ HML_t + b4\ MOM_t + e_it$$

For each fund with at least 36 months of data, I estimate the Carhart (1997) four-factor model in equation (3) using the developed-market factors of Fama and French (2012). The intercept, multiplied by 12, is the fund's annual alpha. One practical point: fund returns are in euros, but the factors are in US dollars. Since the euro weakened against the dollar for much of the period, this pushes all alphas up a little. It matters much less for comparing groups, and the domicile and regional controls soak up a good part of the currency effect.

$$(4)\quad Alpha_i = c + g \times AI_i + d'\ Controls_i + u_i$$

Equation (4) then relates each fund's alpha to its AI intensity (H5, Article 8 and 9 funds only) or to its ESG label (H6, all funds). The controls are fund size in January 2019, fund age, the expense ratio (H5 only), whether the fund is an ETF, the Article 9 label, and domicile in Luxembourg or Ireland. For H5, standard errors are clustered by management company, since funds from the same firm tend to move together; for H6 they are heteroskedasticity-robust. As a second check on H6, I build equal-weighted portfolios of ESG and Article 6 funds and regress the monthly difference in their returns on the four factors.

5.4 Flows (H7)

$$(5)\quad Flow_it = b1\ ESG_i + b2\ Art9_i + b3\ R12_i,t\text{-}1 + b4\ lnTNA_i,t\text{-}1 + b5\ Age_it + m_t + e_it$$

$$(6)\quad Flow_it = c1\ (ESG_i \times Post_t) + c2\ (Art9_i \times Post_t) + b3\ R12_i,t\text{-}1 + b4\ lnTNA_i,t\text{-}1 + f_i + m_t + e_it$$

Equation (5) compares flows across labels within the same month, using month fixed effects, and controls for the previous 12 months' return, fund size and age. Equation (6) adds a fixed effect for each fund, so the label effects come only from changes after 2021 (Post = 1 from January 2022). This

removes anything about a fund that does not change over time. I run the same two models within ESG funds, with the AI Intensity Index in place of the labels.

5.5 Emissions of the ESG-data layer (H8)

(7) *Attributable emissions_t = (Scope 1 + 2 + 3)_t x (Sustainability & Climate revenue_t / Total revenue_t)*

MSCI does not split its emissions by business line, so equation (7) assigns a share of its total emissions to the Sustainability & Climate business in line with that business's share of revenue. I also compare emissions intensity, in tonnes of CO2e per million of revenue, across the five providers.

5.6 Robustness design

Chapter 8 checks the main results in several ways: three-factor instead of four-factor alphas, two separate sub-periods, capped alphas, different standard errors, fixed effects for each fund's regional focus, a placebo test in which the AI score or ESG label is shuffled randomly 1,000 times, and propensity-score matching (Rosenbaum and Rubin, 1983), which pairs each fund with the most similar fund from the comparison group.

6. Results I: Industry and Country Evidence

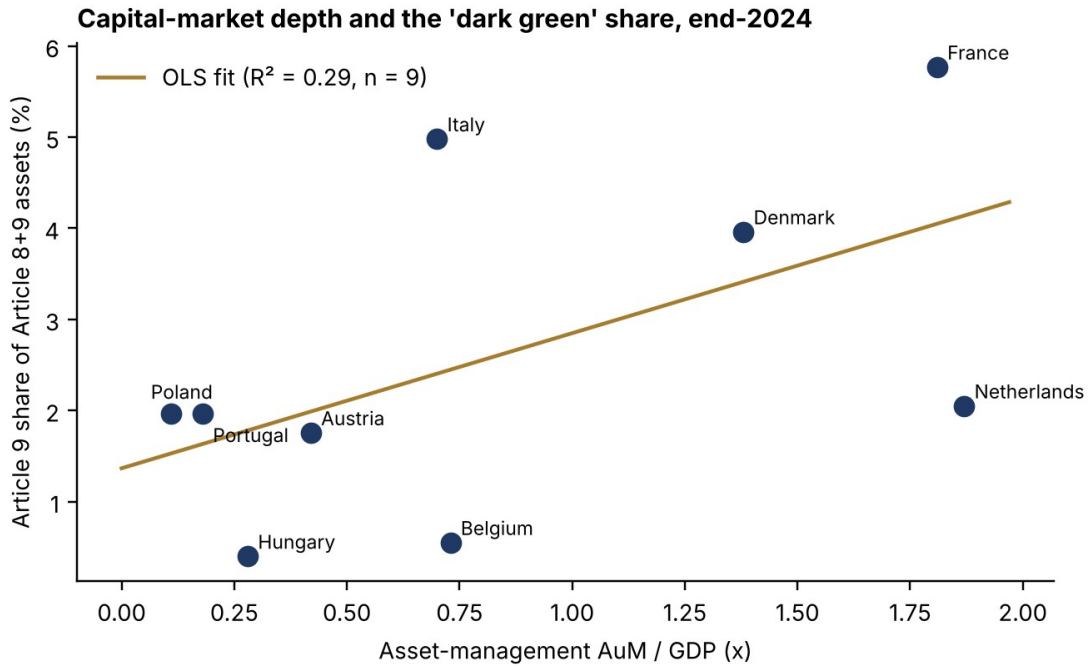
6.1 H1: market depth and the Article 9 share

Among the nine EU countries that report Article 9 assets, deeper markets do tend to hold a larger dark-green share (Table 6.1, Figure 6.1). The correlation is 0.54 and the slope suggests about 1.5 percentage points more Article 9 for each extra unit of AuM/GDP. But the relationship is not statistically significant (p = 0.13), and it leans heavily on France: dropping any single country moves the p-value anywhere between 0.04 and 0.45. So H1 is not supported. The direction fits, but nine countries are simply too few to tell a real pattern apart from the influence of one or two markets. Italy and the Netherlands make the point well: Italy has a 5.0% Article 9 share with AuM at 0.70 times GDP, while the Netherlands has 2.0% with 1.87 times GDP. Pension structures and local demand seem to matter at least as much as depth.

Table 6.1: H1: capital-market depth and the Article 9 share, end-2024

Test	r / R2	p-value	Slope
Pearson correlation	0.541	0.133	
Spearman rank correlation	0.500	0.170	
OLS slope (pp per 1.0x AuM/GDP)	R2 = 0.292	0.133	1.48 (SE 0.87)
Bootstrap 95% CI for the slope (5,000 draws)			[0.02, 3.12]
Excluding France	0.311	0.453	0.78
Excluding Netherlands	0.736	0.037	2.43
Excluding Italy	0.654	0.078	1.58
Excluding Denmark	0.501	0.206	1.38
Excluding Belgium	0.568	0.141	1.43
Excluding Austria	0.524	0.183	1.45
Excluding Poland	0.538	0.169	1.59
Excluding Hungary	0.477	0.232	1.23
Excluding Portugal	0.535	0.172	1.56

Notes: n = 9 EU countries with reported Article 9 assets. Dependent variable: Article 9 share of Article 8+9 fund assets (%).



Sources: EFAMA, Asset Management in Europe (2023, 2025 editions); MSCI SASB tables; Bloomberg.

Figure 6.1: Asset-management AuM/GDP and the Article 9 share of SFDR-labelled fund assets, end-2024.

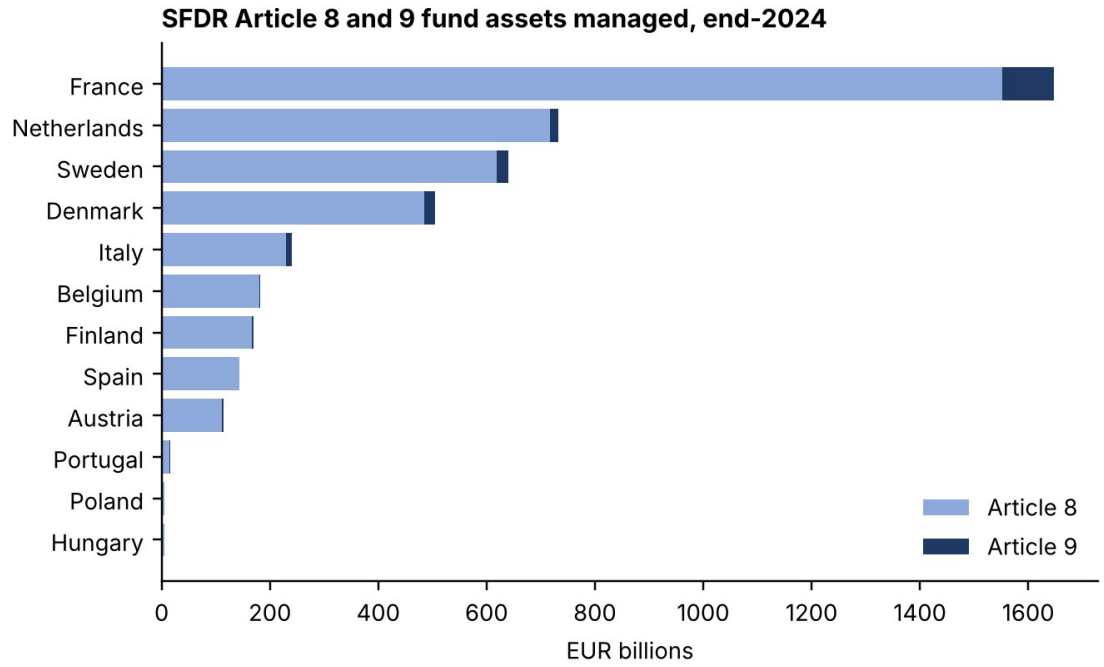
6.2 H2: ESG fund assets scale with market size

Across fifteen EU countries, Article 8 and 9 fund assets rise almost one-for-one with the size of the national industry. The elasticity is 1.19 and the model explains 88% of the variation (Table 6.2). I cannot reject one-for-one scaling (p = 0.14), although the estimate hints that bigger markets hold slightly more than their share. There is a clearer pattern in the related test: the ESG share of a country's assets rises with market depth, so deeper markets are not just bigger but proportionally greener. In the median country, about a quarter of assets (24.7%) sit in Article 8 or 9 funds. H2 holds. Figure 6.2 also shows how concentrated the market is, with France alone managing around EUR 1.6 trillion in Article 8 and 9 funds.

Table 6.2: H2: scaling of SFDR-labelled fund assets with national asset-management size, end-2024

Statistic	Estimate	SE	p-value
Elasticity of Article 8+9 assets to total AuM	1.190	0.119	<0.001
Test: elasticity = 1 (proportional scaling)	t = 1.59		0.135
R-squared	0.885		
Elasticity of the ESG share of AuM to AuM/GDP	0.490	0.197	0.027
Median Article 8+9 assets as % of total AuM	24.7%		

Notes: OLS in logs, n = 15 EU countries. Article 9 assets are added where EFAMA reports them. An elasticity above 1 means ESG fund assets are proportionally more concentrated in the largest asset-management markets.



Sources: EFAMA, Asset Management in Europe (2023, 2025 editions); MSCI SASB tables; Bloomberg.

Figure 6.2: SFDR Article 8 and 9 fund assets managed by country, end-2024 (EUR billions).

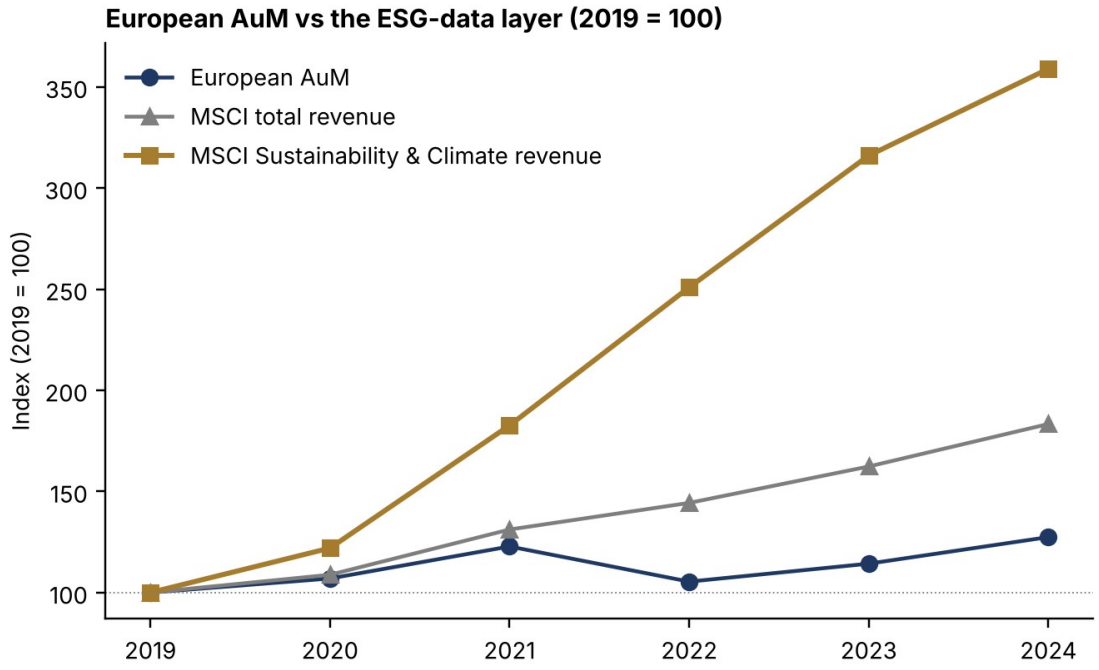
6.3 H3: the ESG-data layer is outgrowing the industry

European assets under management rose from EUR 25.9 trillion in 2019 to EUR 33.0 trillion in 2024, about 5% a year, and fell by 14% in 2022 along the way (Table 6.3, Figure 6.3). MSCI's total revenue grew 13% a year over the same period, and its Sustainability & Climate revenue grew 29% a year, from USD 91m to USD 327m. The segment's share of MSCI's revenue roughly doubled, from 5.8% to 11.4%, and it kept growing by more than a third even in 2022, the year European assets shrank. MSCI's full-time headcount, for comparison, grew 15% a year between 2020 and 2024. So the ESG-data business grew about six times faster than the assets it serves, which supports H3. Two caveats: MSCI is a global company, so some of this growth comes from outside Europe, and revenue tells us how big the business is, not how much computing power it uses.

Table 6.3: H3: European AuM and the ESG-data layer (MSCI), 2019-2024

Year	European AuM (EUR tn)	AuM index (2019=100)	MSCI FT employees	MSCI revenue (USD m)	Sustainability & Climate revenue (USD m)	S&C index (2019=100)	S&C share of revenue
2019	25.9	100	n/a	1,558	91	100	5.8%
2020	27.7	107	3,521	1,695	111	122	6.5%
2021	31.8	123	4,268	2,044	166	183	8.1%
2022	27.3	105	4,722	2,249	228	251	10.2%
2023	29.6	114	5,752	2,529	288	316	11.4%
2024	33.0	127	6,095	2,856	327	359	11.4%
CAGR 2019-2024	5.0%		14.7% (2020-24)	12.9%	29.1%		

Notes: European AuM from EFAMA (2025) Exhibit 1.1 (year-end, includes market valuation effects). MSCI headcount from its SASB tables (2020 as at 30 September). Segment revenue comes from MSCI's segment reporting via Bloomberg; the segment was called 'ESG and Climate' before 2023. MSCI is a global firm, so its growth also reflects non-European demand.



Sources: EFAMA, Asset Management in Europe (2023, 2025 editions); MSCI SASB tables; Bloomberg.

Figure 6.3: European AuM, MSCI total revenue and MSCI Sustainability & Climate revenue, 2019 = 100.

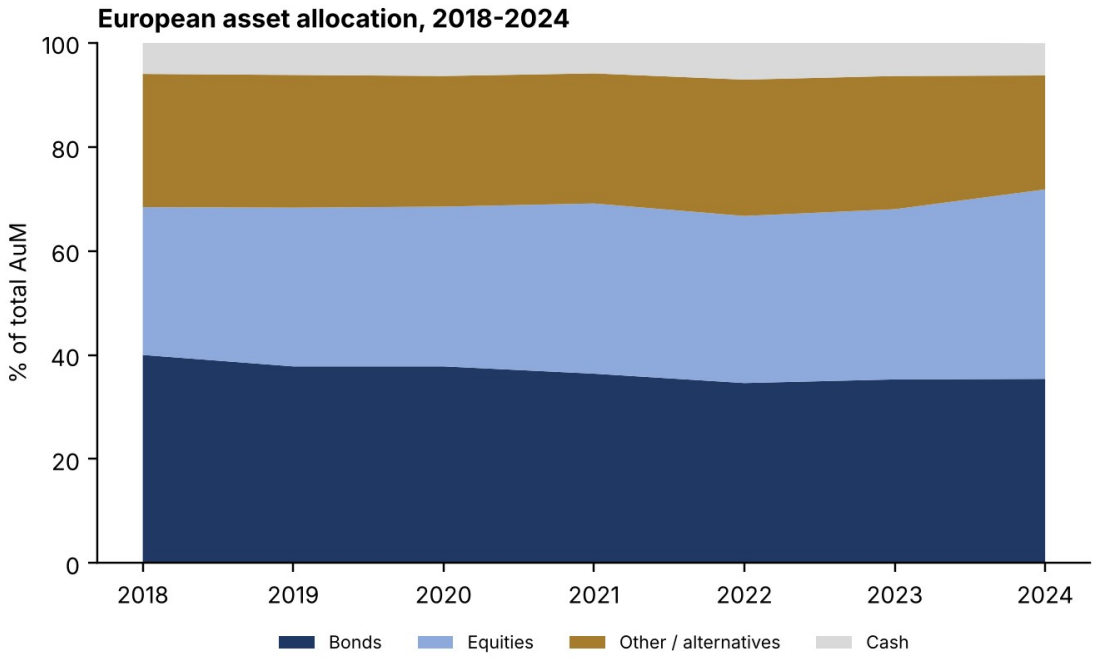
6.4 H4: a shift from bonds to equities

Equities went from 28.4% of European assets in 2018 to 36.4% in 2024, gaining about 1.1 percentage points a year, while bonds fell from 40.0% to 35.4% (Table 6.4, Figure 6.4). Both trends are statistically significant, and 2024 is the first year in which equities overtook bonds. H4 holds. This matters for the wider argument, because owning shares means owning a direct slice of a company's emissions, so the average European portfolio has become more emissions-exposed for reasons that have nothing to do with AI.

Table 6.4: H4: European asset allocation, 2018-2024

Asset class	2018	2024	Trend (pp/year)	R2	p-value
Bonds	40.0%	35.4%	-0.79	0.806	0.006
Equities	28.4%	36.4%	+1.06	0.852	0.003
Other / alternatives	25.6%	21.9%	-0.35	0.285	0.217
Cash / money market	6.0%	6.2%	+0.06	0.111	0.465

Notes: OLS of each asset class share of total European AuM on a linear trend, n = 7 years. EFAMA (2023) Exhibit 4.2 for 2018-2022 and EFAMA (2025) Exhibit 4.2 for 2023-2024; the overlapping years 2020-2022 agree in both editions.



Sources: EFAMA, Asset Management in Europe (2023, 2025 editions); MSCI SASB tables; Bloomberg.

Figure 6.4: European asset allocation by asset class, 2018-2024 (% of total AuM).

6.5 Context: data-centre emissions

Table 6.5 summarises the IEA's estimates. Data centres account for a small share of global emissions today, around 0.5%, but they are one of the few sources expected to keep growing through 2030.

Table 6.5: The scale of data-centre emissions (IEA, Energy and AI, 2025)

Metric	Value
Global fuel-combustion CO2 emissions, 2024	35,000 Mt
Data-centre indirect CO2 emissions, 2024	180 Mt (0.5%)
Data-centre share by 2030, Base Case	1.0%
Data-centre share by 2030, Lift-Off Case	1.4% (emissions about 2.5x)
Potential reductions from wider AI adoption, 2035	up to 1,400 Mt (not on track)

Notes: The emissions multiple and the share multiple differ because total combustion emissions also change by 2030.

7. Results II: Fund-Level Evidence

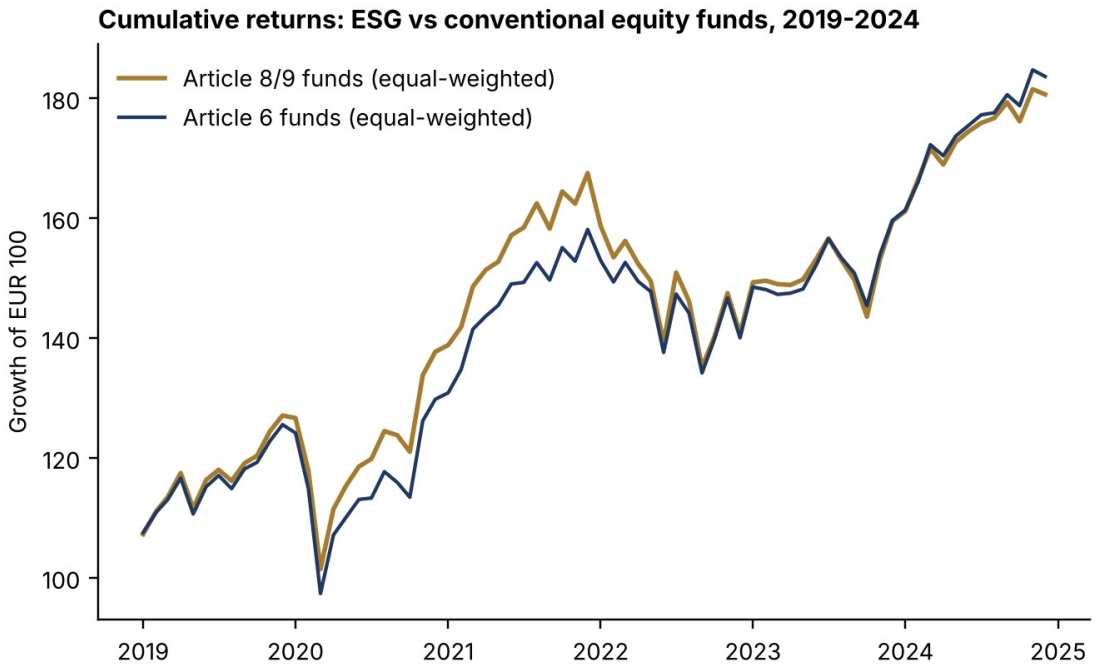
7.1 Performance across SFDR groups

Table 7.1 shows four-factor alphas for 3,899 funds. The averages are 1.35% a year for Article 6, 1.04% for Article 8 and 1.16% for Article 9 funds. That all three are positive reflects the euro-dollar effect explained in Section 5.3, and it applies equally to every group. Only about 2% of funds have alphas that are significantly positive, which is roughly what luck alone would produce. Figure 7.1 shows how closely the two portfolios tracked each other: ESG funds pulled ahead in 2020 and 2021 and then gave the lead back in 2022.

Table 7.1: Risk-adjusted performance by SFDR group: Carhart (1997) four-factor alphas

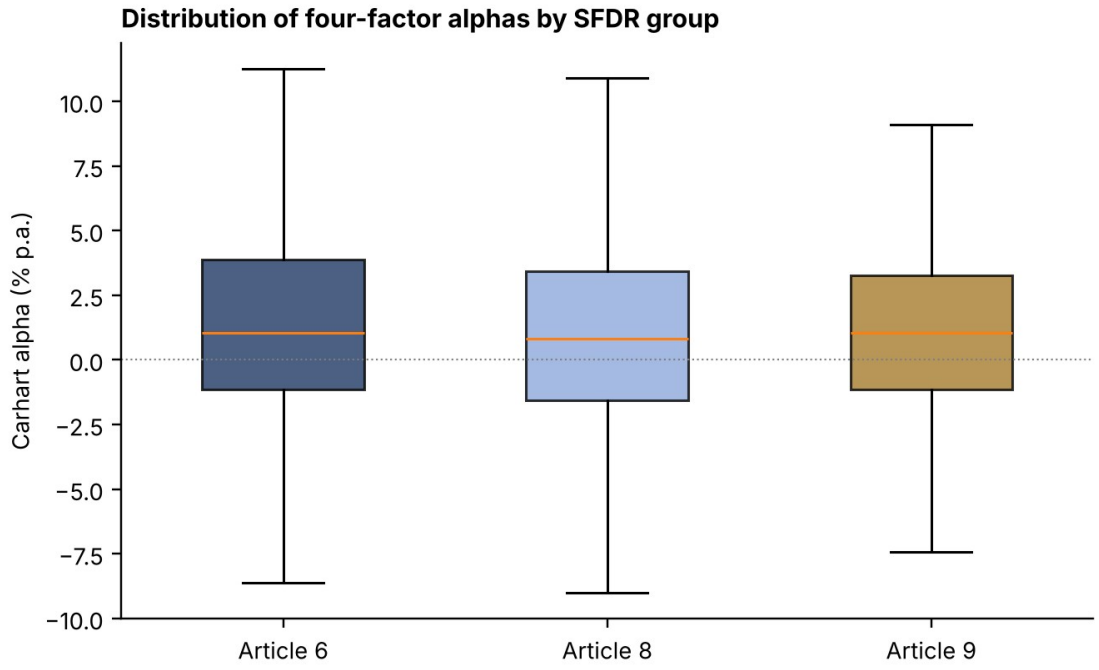
Group	Funds	Mean alpha (% p.a.)	Median alpha	Share alpha t > 1.96	Share alpha t < -1.96	Mean Sharpe	Mean market beta
Article 6	1,455	1.35	1.04	2.1%	0.8%	0.51	0.73
Article 8	2,286	1.04	0.81	1.9%	1.1%	0.51	0.73
Article 9	158	1.16	1.05	0.0%	0.6%	0.50	0.77
Equal-weighted portfolio: Article 8/9 minus Article 6		-0.29	t = -0.54				0.01
Equal-weighted portfolio: Article 9 minus Article 8		0.16	t = 0.16				0.04

Notes: Each fund's monthly excess return in euros is regressed on the developed-market Fama-French factors and momentum (funds with at least 36 months). Alphas annualised (x12). The portfolio rows regress the monthly return difference between equal-weighted portfolios on the same factors (72 months). Source: Bloomberg (fund data, pulled 2 Nov 2026); Kenneth R. French Data Library (developed-market factors).



Source: Bloomberg (fund data, pulled 2 Nov 2026); Kenneth R. French Data Library (developed-market factors).

Figure 7.1: Growth of EUR 100 in equal-weighted portfolios of Article 8/9 and Article 6 equity funds, 2019-2024.



Source: Bloomberg (fund data, pulled 2 Nov 2026); Kenneth R. French Data Library (developed-market factors).

Figure 7.2: Distribution of annualised four-factor alphas by SFDR group (outliers beyond the whiskers not shown).

7.2 H5: AI intensity and performance

At first glance, AI looks like it helps. Within Article 8 and 9 funds, each point on the AI Intensity Index goes with about 0.2 percentage points more alpha a year, which is weakly significant (Table 7.2, column 1). Funds with explicit AI or machine-learning wording seem to earn about 4.5 points more (column 4). But when I looked at who those funds actually are, 14 of the 15 turned out to be funds that invest in AI, robotics and big-data companies, not funds that use AI to pick stocks. Their returns came from owning the technology rally. Once the AI-themed funds are taken out, the effect disappears completely (-0.07, column 5), and the broader measures of data intensity show nothing at any point (columns 2-3). Fees, on the other hand, matter a lot: each percentage point of expense ratio costs about three-quarters of a point of alpha. So the null in H5 stands. There is no reliable sign that using AI or heavy data processing improves returns for ESG investors.

Table 7.2: H5: AI intensity and risk-adjusted performance among Article 8/9 funds

Variable	(1)	(2)	(3)	(4)	(5)
AI Intensity Index (0-10)	0.201*				-0.068
	(1.76)				(-0.65)
Data-intensive (AII > 0)		0.509			
		(1.35)			
AI or quantitative process			0.319		
			(0.72)		
Explicit AI / ML wording				4.495***	
				(3.49)	
ln(AuM), Jan 2019	-0.117*	-0.118*	-0.119*	-0.112*	-0.116*
	(-1.85)	(-1.86)	(-1.87)	(-1.77)	(-1.83)
Fund age (years)	-0.010	-0.010	-0.011	-0.010	-0.009

	(-0.91)	(-0.93)	(-0.98)	(-0.97)	(-0.79)
TER (%)	-0.753***	-0.761***	-0.771***	-0.784***	-0.800***
	(-4.11)	(-4.14)	(-4.19)	(-4.32)	(-4.32)
ETF	-0.639	-0.652	-0.647	-0.753*	-0.798*
	(-1.47)	(-1.50)	(-1.47)	(-1.74)	(-1.94)
Article 9	0.391	0.395	0.388	0.382	0.404
	(1.20)	(1.21)	(1.19)	(1.20)	(1.25)
Luxembourg-domiciled	1.851***	1.854***	1.862***	1.838***	1.832***
	(6.73)	(6.72)	(6.79)	(6.72)	(6.66)
Ireland-domiciled	2.825***	2.822***	2.820***	2.809***	2.787***
	(8.29)	(8.26)	(8.23)	(8.30)	(8.16)
Constant	1.488***	1.515***	1.550***	1.535***	1.560***
	(3.61)	(3.64)	(3.74)	(3.72)	(3.75)
Sample	Art. 8/9	Art. 8/9	Art. 8/9	Art. 8/9	excl. AI-themed
Observations	2,206	2,206	2,206	2,206	2,187
R-squared	0.111	0.110	0.109	0.117	0.106

*Notes: Dependent variable: annualised Carhart alpha (% p.a.). OLS; t-statistics in parentheses based on standard errors clustered by management company. Column (5) drops the funds that invest in the AI or robotics theme (their holdings, not their process, are AI-related). * **p<0.01**, p<0.05, * p<0.10.*

7.3 H6: ESG and conventional funds

Once size, age, ETF status and domicile are taken into account, Article 8 and 9 funds earn essentially the same alpha as Article 6 funds: the difference is -0.05 percentage points a year with a p-value of 0.73 (Table 7.3). The picture is the same for Sharpe ratios and for actively managed funds on their own, and Article 9 funds do no better or worse than Article 8 funds. The portfolio test agrees: ESG minus conventional comes out at -0.29% a year, and Article 9 minus Article 8 at 0.16%, neither of them significant. Over the full six years, then, H6 holds. As Chapter 8 shows, though, that average hides two very different halves.

Table 7.3: H6: Article 8/9 funds versus conventional (Article 6) funds

Variable	(1)	(2)	(3)
Article 8/9 (ESG)	-0.047	0.019	-0.109
	(-0.34)	(1.40)	(-0.74)
Article 9	0.127	-0.002	0.126
	(0.49)	(-0.12)	(0.48)
ln(AuM), Jan 2019	-0.063	0.005	-0.068
	(-1.50)	(1.51)	(-1.47)
Fund age (years)	-0.039***	-0.003***	-0.035***
	(-5.23)	(-4.35)	(-4.56)
ETF	-0.177	-0.003	
	(-0.88)	(-0.18)	
Luxembourg-domiciled	1.446***	0.021	1.603***
	(10.32)	(1.51)	(11.17)
Ireland-domiciled	2.405***	0.103***	2.406***
	(13.02)	(6.56)	(11.57)
Constant	0.889***	0.473***	0.828***
	(3.58)	(22.34)	(3.15)
Dependent variable	Alpha	Sharpe	Alpha
Funds	All	All	Active (no ETFs)

Observations	3,891	3,891	3,215
R-squared	0.078	0.022	0.080

Notes: Cross-section of funds. Alpha = annualised Carhart alpha (% p.a.); Sharpe = annualised Sharpe ratio. HC1-robust t-statistics in parentheses (management company is not available for Article 6 funds).

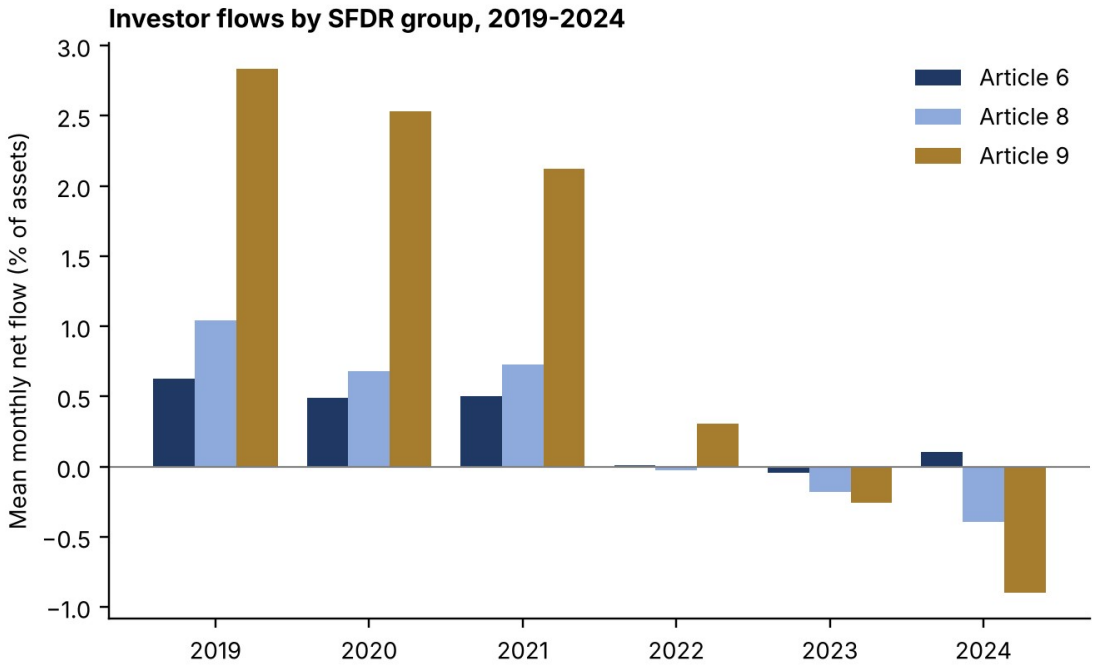
7.4 H7: investor flows

As expected, money follows past returns, and this is the strongest effect in the flow models. Even after allowing for performance, size and age, Article 9 funds took in about 0.5 percentage points of their assets a month more than other funds, and Article 8 funds slightly more than Article 6 (Table 7.4, column 1). That changed after 2021. With fund and month fixed effects, the ESG premium fell by about 0.26 points and the Article 9 premium by a further 0.9 points (column 2). Figure 7.3 shows how sharp the turn was: Article 9 funds gained about 2.8% of their assets a month in 2019 and were losing 0.9% a month by 2024, while Article 6 funds stayed close to zero throughout. The timing matches the 2022 energy shock, the weaker run for ESG funds and the wave of Article 9 downgrades in late 2022. H7 holds. Within ESG funds, AI intensity attracted no extra money on average, but it did pick up a modest premium after 2021, which fits the investor excitement around AI from 2023.

Table 7.4: H7: investor flows, ESG labels and AI intensity (fund-month panel)

Variable	(1)	(2)	(3)	(4)
Article 8/9 (ESG)	0.079*			
	(1.76)			
Article 9	0.540***		0.639***	
	(5.77)		(5.85)	
ESG x 2022-24		-0.262***		
		(-3.18)		
Article 9 x 2022-24		-0.893***		
		(-3.56)		
AI Intensity Index (0-10)			-0.077	
			(-1.60)	
AII x 2022-24				0.171**
				(2.28)
Past 12-month return	0.052***	0.057***	0.074***	0.078***
	(22.68)	(24.76)	(16.87)	(18.31)
ln(AuM), lagged	-0.170***	-1.891***	-0.207***	-1.790***
	(-11.63)	(-23.49)	(-8.82)	(-12.11)
Fund age (years)	-0.036***		-0.027***	
	(-15.80)		(-8.00)	
TER (%)			0.015	
			(0.30)	
ETF			1.045***	
			(4.47)	
Fixed effects	Month	Fund + month	Month	Fund + month
Sample	All funds	All funds	Art. 8/9	Art. 8/9
Observations	230,563	230,563	131,010	131,010
R-squared	0.019	0.030	0.034	0.040

Notes: Dependent variable: monthly net flow, % of lagged assets (Sirri and Tufano, 1998), winsorised at 1%/99%. Post-2021 = 2022-2024. Fixed effects removed by within transformation; t-statistics clustered by fund (columns 1-2) or management company (3-4). Time-invariant fund characteristics are absorbed by fund fixed effects in columns (2) and (4).



Source: Bloomberg (fund data, pulled 2 Nov 2026); Kenneth R. French Data Library (developed-market factors).

Figure 7.3: Mean monthly net flows by SFDR group and year (% of lagged fund assets).

7.5 H8: the footprint of the ESG-data layer

MSCI's total emissions fell from 62.9 thousand tonnes of CO2e in 2019 to 51.5 thousand tonnes in 2024, and its emissions per million dollars of revenue more than halved, from 40 to 18 tonnes (Table 7.5). Nearly all of its footprint, 99% in 2024, is Scope 3, the supply chain that includes the cloud computing and data centres it buys in. All five providers became more carbon-efficient over the period (Table 7.6). But MSCI's ESG business grew so quickly that the emissions belonging to it still rose, from 3.7 to 5.9 thousand tonnes, an increase of 60% (Figure 7.4). H8 holds: the efficiency gains are real, but the business is growing faster. The scale matters here, though. About six thousand tonnes a year for the largest ESG-ratings business is tiny next to the millions of tonnes financed through fund portfolios. The case for disclosure rests on transparency and on the direction of travel, not on the size of the number.

Table 7.5: H8: MSCI emissions and the footprint attributable to its Sustainability & Climate business

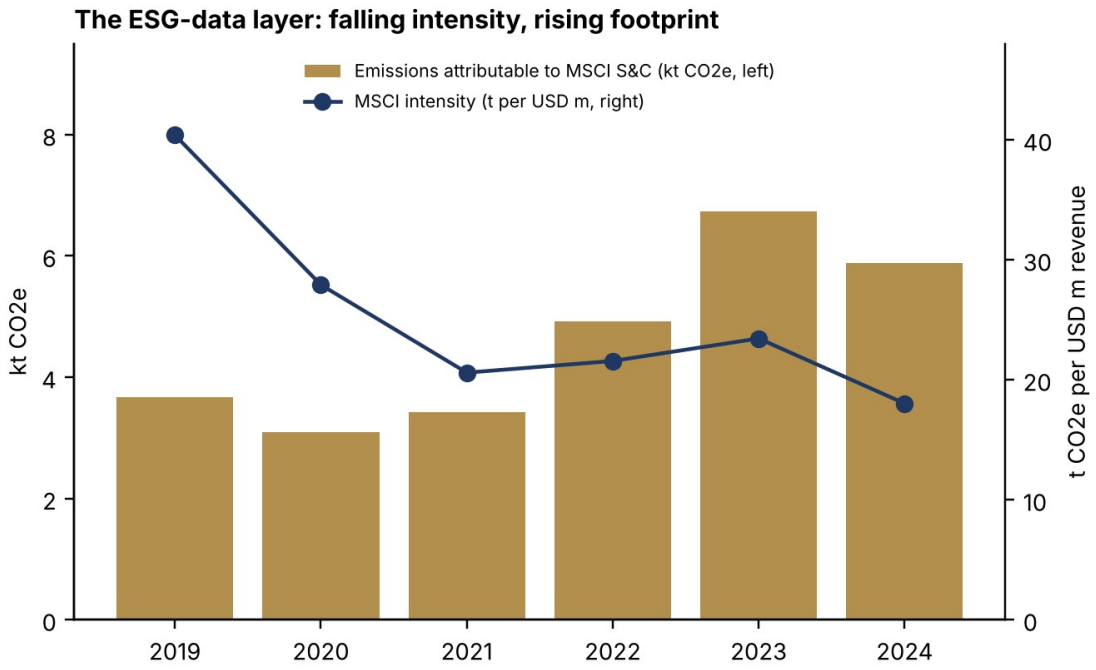
Year	MSCI Scope 1+2 (kt CO2e)	Scope 3 (kt)	Total (kt)	Intensity (t per USD m revenue)	S&C share of revenue	Attributable to S&C (kt)
2019	4.70	58.2	62.9	40.4	5.8%	3.67
2020	3.97	43.3	47.3	27.9	6.5%	3.10
2021	1.35	40.7	42.1	20.6	8.1%	3.42
2022	0.83	47.7	48.5	21.6	10.2%	4.92
2023	0.65	58.6	59.3	23.4	11.4%	6.74
2024	0.64	50.8	51.5	18.0	11.4%	5.89

Notes: Scope 1, Scope 2 (market-based) and Scope 3 emissions in thousand tonnes of CO2e, from Bloomberg. Emissions attributable to S&C = total emissions x the segment's revenue share (a revenue-based allocation, since MSCI does not report emissions by segment).

Table 7.6: H8: emissions intensity of the five largest listed ESG-data providers (tCO2e per million of revenue)

Provider	2019	2024	Change
MSCI	40.4	18.0	-55%
S&P Global	108.1	31.4	-71%
Moody's	42.0	17.6	-58%
LSEG	284.2	46.5	-84%
Morningstar	11.6	4.8	-59%

Notes: Scope 1 + Scope 2 (market-based) + Scope 3 from Bloomberg; revenue in reporting currency (LSEG in GBP, others USD). LSEG's 2021 jump in revenue reflects the Refinitiv acquisition. Scope 3 boundaries differ between companies, so levels are not strictly comparable; changes over time are more informative.



Source: Bloomberg (MSCI emissions and segment revenue).

Figure 7.4: Emissions attributable to MSCI's Sustainability & Climate business (bars) and MSCI's emissions intensity (line).

8. Robustness and Endogeneity

8.1 Robustness

Table 8.1 repeats the main H5 and H6 models, changing one thing at a time. For H5, the small positive AI effect shows up with three-factor alphas and in 2022-2024, but it is zero in 2019-2021. It stops being significant once each fund's regional focus is controlled for, and turns slightly negative when the AI-themed funds are also removed. The placebo test tells me the raw link is not just random noise, and the other rows show where it comes from: exposure to technology stocks during the AI rally. Matching each AI or quantitative fund with a similar fund finds no meaningful difference. The conclusion that using AI does not improve performance holds up.

For H6, the full-period result survives every change. The two halves of the sample are the interesting part: ESG funds beat conventional funds by about 1.2 percentage points a year in 2019-2021 and then fell behind by about 0.4 points in 2022-2024, both clearly significant. That is the pattern Pástor, Stambaugh and Taylor (2022) would predict: green assets do well while worries about climate are building, and lose that edge when an energy price spike and rising rates hit markets. Matching ESG funds with similar conventional funds gives a small negative gap over the full period, so if anything ESG funds lagged slightly overall, but by a small amount that depends heavily on which years you look at.

Table 8.1: Robustness: alternative models, samples and placebo tests

Test	H5: coefficient on AII (t)	H6: coefficient on ESG (t)
Fama-French 3-factor alphas (no momentum)	0.219 (2.22)	-0.079 (-0.57)
Sub-period 2019-2021	0.002 (0.01)	1.151 (5.96)
Sub-period 2022-2024	0.413 (4.06)	-0.421 (-2.44)
Alphas winsorised at 5%/95%	0.158 (1.63)	-0.053 (-0.46)
HC1 standard errors instead of clustering	0.201 (2.08)	as main
Adding geographic-focus fixed effects (ESG funds)	0.139 (1.27)	n.a.
Geographic FE and excluding AI-themed funds	-0.127 (-1.55)	n.a.
Placebo: AII randomly reassigned (1,000 draws)	permutation p = 0.020	permutation p = 0.717
Propensity-score matching: AI/quant process vs matched funds (ATT)	0.265 (t = 0.50; 99 of 99 matched)	-0.335 (t = -3.05; 2,442 of 2,442 matched)

Notes: Each row repeats the main model (Table 7.2, column 1 for H5; Table 7.3, column 1 for H6) with one change. Matching: one-to-one nearest neighbour with replacement on the propensity score (caliper 0.05 for H5, 0.02 for H6). After matching, the largest standardised difference in covariates is 14.2% (H5) and 14.4% (H6).

8.2 Endogeneity

None of these results prove cause and effect. Managers choose their own SFDR label and decide how to describe their strategy, so labels and AI wording may partly reflect skill, marketing or investment style that I cannot see. I deal with this in three ways: detailed controls and regional fixed effects, fund fixed effects in the flow models so that only changes over time count, and propensity-score matching. Some imbalance remains after matching, with the largest standardised difference at about 14%. At country level, deeper markets may attract ESG products for reasons other than depth, such as how pensions are organised or local regulation, and fifteen countries are not enough to separate these.

9. Discussion and Limitations

9.1 Discussion

The evidence supports both halves of the AI-ESG paradox, although in a milder form than the name might suggest. On the cost side, the ESG-data business is growing around six times faster than European assets, and its share of emissions is rising even though providers are becoming much more efficient. On the benefit side, investors get no measurable reward: ESG funds match conventional funds over the full period, and the funds that rely most on data and AI do no better than other ESG funds once you strip out their tech holdings. What investors did respond to was the label itself. Article 9 funds drew heavy inflows until 2021 and lost money after the 2022 shock, which is what signalling theory and Hartzmark and Sussman (2019) would lead you to expect.

The policy point that follows is modest but practical. ESG data are not a major source of emissions, and I do not want to overstate that. The issue is that the costs and claims of the data layer are invisible to the investors who end up paying for them. If ESG products disclosed the data and technology behind them, in the spirit of the SASB materiality framework, investors could judge for themselves whether an AI-heavy ESG strategy is worth what it costs.

9.2 Limitations

- **Survivorship.** The screen only captures funds that were still active in 2026, so closed and merged funds are missing from all three groups.
- **Currency.** Fund returns are in euros and the factors in US dollars. This raises the level of all alphas but has much less effect on comparisons between groups.
- **Measuring AI.** The index is built from short Bloomberg descriptions, so it reflects what funds say rather than what they do, and only 138 ESG funds score above zero.
- **Current labels.** SFDR labels are as of 2026, so funds that moved from Article 9 to Article 8 in 2022-2023 appear under their current label.
- **Article 6 detail.** Expense ratios and descriptions were only collected for Article 8 and 9 funds, so the H6 comparison cannot control for fees.
- **Emissions allocation.** Emissions are split by revenue share, and companies draw their Scope 3 boundaries differently.
- **Small country samples.** H1 is based on nine countries and H2 on fifteen.

10. Conclusion and Future Research

I set out to answer three questions: is the data and AI infrastructure behind European ESG investing growing faster than the assets it serves, does it help investors, and what does it cost in emissions? Using EFAMA's industry statistics, a Bloomberg panel of 3,917 equity funds and the disclosures of the five largest ESG-data providers, I find that MSCI's ESG business grew by about 29% a year against 5% for European assets. ESG funds earned the same risk-adjusted returns as conventional funds overall, doing better in 2019-2021 and worse in 2022-2024. AI-intensive ESG funds showed no advantage once their thematic holdings were set aside. Investors rewarded the Article 9 label until 2021 and pulled back from 2022, and the emissions linked to the ESG-data business rose by 60% even as every provider became more carbon-efficient. These findings make a reasonable case for including the footprint of the ESG-data supply chain in ESG disclosure.

Future research

- Using holdings-based fund emissions (weighted average carbon intensity) to compare the data-layer footprint directly with portfolio emissions.
- Tracking SFDR reclassification histories to see how Article 9 downgrades affected flows and performance.
- Measuring AI use directly, for example through managers' job postings, patents or technology spending, instead of fund descriptions.
- Repeating the analysis with survivorship-free data that includes closed and merged funds.

References

Akerlof, G.A. (1970). The market for 'lemons': Quality uncertainty and the market mechanism. Quarterly Journal of Economics, 84(3), 488-500.

Berg, F., Kölbel, J.F. and Rigobon, R. (2022). Aggregate confusion: The divergence of ESG ratings. Review of Finance, 26(6), 1315-1344.

Bloomberg L.P. (2026). Bloomberg Professional service: fund screening, fund data, company ESG data and financial analysis. Data retrieved 2 November 2026, DCU Bloomberg lab.

Carhart, M.M. (1997). On persistence in mutual fund performance. Journal of Finance, 52(1), 57-82.

Chatterji, A.K., Durand, R., Levine, D.I. and Touboul, S. (2016). Do ratings of firms converge? Implications for managers, investors and strategy researchers. Strategic Management Journal, 37(8), 1597-1614.

EFAMA (2023). Asset Management in Europe: An overview of the asset management industry, 15th edition. Brussels: European Fund and Asset Management Association.

EFAMA (2025). Asset Management in Europe: An overview of the asset management industry, 17th edition. Brussels: European Fund and Asset Management Association.

Fama, E.F. and French, K.R. (1993). Common risk factors in the returns on stocks and bonds. Journal of Financial Economics, 33(1), 3-56.

Fama, E.F. and French, K.R. (2012). Size, value, and momentum in international stock returns. Journal of Financial Economics, 105(3), 457-472.

Fisch, J.E., Labouré, M. and Turner, J.A. (2019). The emergence of the robo-advisor. Pension Research Council Working Paper 2019-12, University of Pennsylvania.

Freeman, R.E. (1984). Strategic Management: A Stakeholder Approach. Boston: Pitman.

French, K.R. (2026). Data Library: Developed market factors and returns. Tuck School of Business, Dartmouth College.

Friede, G., Busch, T. and Bassen, A. (2015). ESG and financial performance: Aggregated evidence from more than 2000 empirical studies. Journal of Sustainable Finance & Investment, 5(4), 210-233.

Hartzmark, S.M. and Sussman, A.B. (2019). Do investors value sustainability? A natural experiment examining ranking and fund flows. Journal of Finance, 74(6), 2789-2837.

International Energy Agency (2025). Energy and AI: AI and climate change. Paris: IEA.

Kotsantonis, S. and Serafeim, G. (2019). Four things no one will tell you about ESG data. Journal of Applied Corporate Finance, 31(2), 50-58.

Lacoste, A., Luccioni, A., Schmidt, V. and Dandres, T. (2019). Quantifying the carbon emissions of machine learning. NeurIPS 2019 Workshop on Tackling Climate Change with Machine Learning.

López de Prado, M. (2018). Advances in Financial Machine Learning. Hoboken: Wiley.

MSCI Inc. (2021-2025). SASB Reference Tables, reporting years 2020-2024.

Our World in Data (2025). CO2 emissions. Data from the Global Carbon Budget (2025).

Pástor, L., Stambaugh, R.F. and Taylor, L.A. (2021). Sustainable investing in equilibrium. Journal of Financial Economics, 142(2), 550-571.

Pástor, L., Stambaugh, R.F. and Taylor, L.A. (2022). Dissecting green returns. Journal of Financial Economics, 146(2), 403-424.

Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L., Rothchild, D., So, D., Texier, M. and Dean, J. (2021). Carbon emissions and large neural network training. arXiv:2104.10350.

Pedersen, L.H., Fitzgibbons, S. and Pomorski, L. (2021). Responsible investing: The ESG-efficient frontier. Journal of Financial Economics, 142(2), 572-597.

Raghunandan, A. and Rajgopal, S. (2022). Do ESG funds make stakeholder-friendly investments? Review of Accounting Studies, 27(3), 822-863.

Renneboog, L., Ter Horst, J. and Zhang, C. (2008). Socially responsible investments: Institutional aspects, performance, and investor behavior. Journal of Banking & Finance, 32(9), 1723-1742.

Riedl, A. and Smeets, P. (2017). Why do investors hold socially responsible mutual funds? Journal of Finance, 72(6), 2505-2550.

Rosenbaum, P.R. and Rubin, D.B. (1983). The central role of the propensity score in observational studies for causal effects. Biometrika, 70(1), 41-55.

Schwartz, R., Dodge, J., Smith, N.A. and Etzioni, O. (2020). Green AI. Communications of the ACM, 63(12), 54-63.

Sirri, E.R. and Tufano, P. (1998). Costly search and mutual fund flows. Journal of Finance, 53(5), 1589-1622.

Spence, M. (1973). Job market signaling. Quarterly Journal of Economics, 87(3), 355-374.

Strubell, E., Ganesh, A. and McCallum, A. (2019). Energy and policy considerations for deep learning in NLP. Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 3645-3650.

Appendix A: Data and Code Availability

The analysis behind this dissertation uses four data files and one Python script. They are not published here, but are available on request: please contact me on LinkedIn (linkedin.com/in/suyashrch) or at suyashrch@gmail.com.

Data files:

- **Industry Dataset.xlsx**: EFAMA country and industry statistics, MSCI staff and segment revenue, ESG-data provider emissions and IEA figures, with a sheet listing the source of every series (Chapter 6 and Section 7.5).
- **Fund Characteristics.csv**: the 3,917 Europe-domiciled equity funds with their SFDR classification, fund type, domicile, launch date, size, expense ratio, management company, benchmark, regional focus and Bloomberg description (Section 4.3).
- **Fund Monthly Data.csv**: monthly total return (%), net asset value and total assets (EUR m) for every fund, January 2019 to December 2024 (280,865 fund-months).
- **Risk Factors.csv**: the developed-market Fama-French factors and momentum used for the alphas, % per month.

The analysis code is a single Python file, **Suyash Sawant AI-ESG Analysis.py**. Placed in the same folder as the four data files, it builds the AI Intensity Index and the fund panel and reproduces every table and figure in this dissertation. It needs Python 3 with pandas, NumPy, SciPy, Matplotlib and openpyxl.

The fund data were downloaded from Bloomberg at the DCU Bloomberg lab on 2 November 2026 and are subject to Bloomberg's licence terms, so they can be shared for academic and review purposes only. The factor data come from the freely available Kenneth R. French Data Library.

Appendix B: Variable Definitions

Table B.1: Variable definitions

Variable	Definition	Source
Alpha	Intercept from the Carhart four-factor regression of monthly excess returns, multiplied by 12 (funds with at least 36 months)	Bloomberg; French
Sharpe ratio	Average monthly excess return divided by its standard deviation, annualised	Bloomberg; French
Flow	[TNA(t) - TNA(t-1)(1 + R(t))] / TNA(t-1), % per month, top and bottom 1% capped	Bloomberg
ESG / Article 9	1 if the fund is classified under SFDR Article 8 or 9 / Article 9	Bloomberg
AII	AI Intensity Index, 0-10 (Appendix C)	Bloomberg descriptions
ln(AuM)	Natural log of fund assets (EUR m), January 2019 or the previous month	Bloomberg
Age	Years since launch	Bloomberg
TER	Total expense ratio (%)	Bloomberg
2022-24	1 from January 2022 onwards	-
Attributable emissions	Provider Scope 1+2+3 emissions x the segment's share of revenue	Bloomberg

Notes: All returns are total returns in euros with dividends reinvested.

Appendix C: AI Intensity Index Dictionary

Table C.1: AI Intensity Index: keyword dictionary

Category	Terms searched in fund name and Bloomberg description	Article 8/9 funds flagged
AI / machine learning (weight 3)	AI, artificial intelligence, machine learning, deep learning, neural network, natural language, big data, data science, alternative data	15
Quantitative / algorithmic process (weight 2)	quantitative, quant, systematic, algorithm(ic), model-driven, model-based, proprietary model, QI	90
Rules-based / factor / automation (weight 1)	rules-based, smart beta, multi-factor, factor, enhanced index, optimised, robotics, automation	50

Notes: AII = 10 x (sum of the weights of the categories present) / 6, so it ranges from 0 to 10. Matching is case-insensitive with word boundaries for short terms. AI-themed funds (holdings in AI, robotics or big-data companies) are flagged separately for robustness.